

CAGAYAN DE ORO, Philippines

WMO: 987480

Lat: 8.484N

Lon: 124.647E

Elev: 20

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 86-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.8	73.0	65.0	92.6	74.8	66.9	99.1	75.9	13.1	81.8	11.3	86.5	2.1	220	0.316

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.9	94.3	81.5	93.3	81.3	92.5	81.1	83.8	91.7	83.2	91.1	82.6	90.6	4.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.7	164.9	90.2	80.9	160.7	89.8	80.2	157.0	89.2	48.0	91.8	47.2	91.4	46.5	90.8	91.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	8.4	6.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.8	80.6	81.3	82.1	83.4	85.0	83.5	83.1	83.7	83.3	82.9	82.4	81.6
	DBStd	2.11	2.00	1.89	1.79	1.73	1.70	1.78	1.68	1.89	1.65	1.48	1.80	1.85
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11957	950	877	995	1003	1085	1004	1026	1044	1000	1021	973	979
	CDD65	6482	485	457	530	553	620	554	561	579	550	556	523	514
	CDH74	70286	4563	4503	5710	6526	7770	6118	5971	6531	5932	6054	5458	5150
	CDH80	28358	1472	1591	2255	2826	3676	2515	2404	2784	2432	2437	2118	1848
Wind	WSAvg	2.4	2.6	2.3	2.6	2.5	2.5	2.1	2.3	2.2	2.3	2.1	2.4	2.4
Precipitation	PrecAvg	65.6	4.7	3.4	2.0	1.3	4.1	8.5	7.9	9.0	8.7	7.0	5.4	3.7
	PrecMax	80.4	17.8	7.1	6.3	3.1	9.1	14.1	16.3	13.0	14.7	12.4	14.6	9.5
	PrecMin	43.3	0.0	0.2	0.1	0.0	0.5	2.4	1.8	4.3	2.2	2.3	0.9	0.4
	PrecStd	10.1	5.0	2.4	1.7	0.9	2.8	3.1	4.1	2.7	3.6	3.0	3.8	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.5	91.9	93.4	93.6	95.8	94.4	94.1	95.8	94.2	93.4	93.1	92.1
		MCWB	79.7	79.1	79.9	82.0	81.3	82.0	80.9	82.1	81.6	81.4	80.4	80.9
	2%	DB	89.9	90.5	91.7	92.5	94.2	92.6	92.4	93.4	92.7	92.0	91.6	90.5
		MCWB	79.3	79.1	79.6	81.3	81.5	81.2	81.5	81.3	81.1	81.2	80.5	80.6
	5%	DB	88.7	89.3	90.5	91.4	93.1	91.5	91.2	91.9	91.5	90.9	90.5	89.4
		MCWB	79.1	79.0	79.7	80.7	81.5	81.0	81.0	81.0	80.9	81.1	80.3	80.2
	10%	DB	87.2	88.0	89.1	90.4	91.9	90.3	89.9	90.6	90.2	89.7	89.2	88.0
		MCWB	78.7	78.8	79.6	80.5	81.2	80.6	80.4	80.7	80.5	80.8	80.0	79.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.1	81.9	83.3	84.1	84.5	84.3	84.2	84.0	83.6	83.7	83.3	82.7
		MCDB	89.0	89.2	90.2	92.1	92.7	92.2	91.5	92.5	91.9	91.1	90.4	89.3
	2%	WB	80.9	80.9	81.8	82.8	83.5	83.0	82.9	82.8	82.6	82.6	82.2	81.7
		MCDB	87.7	88.3	89.2	90.6	91.9	91.1	90.7	91.3	90.8	90.3	89.5	88.7
	5%	WB	80.0	80.1	81.0	81.9	82.6	82.0	82.0	81.9	81.8	81.7	81.3	80.8
		MCDB	86.7	87.5	88.7	89.7	91.2	90.2	89.9	90.5	89.9	89.6	88.8	88.1
	10%	WB	79.1	79.2	80.1	81.2	81.8	81.0	81.0	81.0	80.9	80.9	80.3	80.0
		MCDB	85.8	86.6	87.8	89.0	90.4	88.9	88.7	89.6	89.0	88.9	87.9	87.1
Mean Daily Temperature Range	5% DB	MDBR	12.9	13.5	14.6	14.8	14.9	14.3	14.6	14.9	14.9	14.6	14.6	14.1
		MCDBR	15.7	15.8	16.6	16.0	16.0	15.7	16.1	16.4	16.3	16.1	16.3	15.7
		MCWBR	8.9	8.7	9.5	8.9	8.0	8.1	8.8	8.7	8.8	8.9	9.0	8.9
	5% WB	MCDBR	14.3	14.7	15.2	15.2	15.3	14.7	15.4	15.4	15.7	15.2	14.8	15.1
MCWBR		9.0	8.9	9.7	9.3	8.9	8.6	9.4	9.0	9.2	9.1	9.1	9.3	
Clear-Sky Solar Irradiance	taub	0.408	0.404	0.406	0.417	0.426	0.417	0.416	0.438	0.437	0.444	0.414	0.408	
	taud	2.490	2.482	2.484	2.450	2.440	2.471	2.470	2.393	2.390	2.368	2.472	2.487	
	Ebn at Noon	287	291	290	283	275	274	275	273	277	276	283	285	
	Edh at Noon	35	36	36	37	36	35	35	39	39	40	35	34	
All-Sky Solar Radiation	RadAvg	1303	1534	1733	1859	1728	1596	1576	1641	1627	1562	1456	1363	
	RadStd	171	171	135	103	104	77	85	108	107	100	125	120	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.40	N/A	N/A	+0.79	+0.59	+0.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page